

DIGITAL FORENSICS EXAMINATION REPORT

Case Name:	Operation Dark Web
Case Number:	CY-2024-0090
Examining Analyst:	Agent K. Patel
Department:	Digital Forensics Lab
Date of Report:	May 28, 2026 11:04 UTC
Classification:	CONFIDENTIAL — FOR OFFICIAL USE ONLY

SUMMARY STATISTICS

Artifact Category	Count	Artifact Category	Count
Total Artifacts	15	Normalized Events	0
Web History	4	Emails	2
USB Devices	1	Deleted Files	2
User Accounts	1	Search Queries	2
High Confidence	0	Suspicious Events	3

1. EXECUTIVE SUMMARY

This report presents findings of a digital forensic examination of Operation Dark Web (Case No. CY-2024-0090), conducted by Agent K. Patel of the Digital Forensics Lab.

The examination yielded 15 digital artifacts including 4 web history entries, 2 email records, 1 USB device connections, and 2 recovered deleted files. High-confidence events: 0. Suspicious events: 3.

All evidence has been handled per chain of custody protocols to maintain admissibility.

2. METHODOLOGY

The digital forensic examination was conducted in accordance with NIST SP 800-86 and ACPO Good Practice Guide for Digital Evidence.

- Evidence Acquisition: Subject digital media was forensically processed using write-blocking procedures. MD5 hash values recorded to verify integrity.
- Evidence Verification: Forensic image verified for accuracy. File hash recorded in chain of custody documentation.
- Artifact Extraction: Evidence processed through automated parsing pipeline covering web history, email, USB devices, user accounts, deleted files, registry entries, installed programs, and search queries.
- Normalization: All artifacts normalized into unified schema with full traceability: source file, row/line reference, timestamp, artifact ID, and confidence rating (High/Medium/Low).
- RAG Indexing: All artifacts indexed for cross-referencing and AI-assisted analysis. Each AI finding cites the exact source artifact(s) it is based on.
- Documentation: All findings documented with evidence references in format: 'Finding: [X] | Evidence: [filename] (Row [N]) | Time: [T] | Artifact: [ID]'.

3. KEY FINDINGS

3.1 Web Activity

Examination revealed 4 web browsing history entries providing a chronological record of websites visited, indicating user intent relevant to the investigation.

Finding: URL visited: {'Artifact Type': 'TSK_WEB_HISTORY', 'Path': 'https://bitcointalk.org/', 'Date/Time': '2024-04-01 22:44:10'}

Evidence: [operation_dark_web_sample.csv \(Row 5\)](#) | Time: 2024-04-01 22:44:10 | Artifact: web_5 | Confidence: High

Finding: URL visited: {'Artifact Type': 'TSK_WEB_HISTORY', 'Path': 'https://www.torproject.org/download/', 'Date/Time': '2024-04-01 23:58:01'}

Evidence: [operation_dark_web_sample.csv \(Row 3\)](#) | Time: 2024-04-01 23:58:01 | Artifact: web_3 | Confidence: High

Finding: URL visited: {'Artifact Type': 'TSK_WEB_HISTORY', 'Path': 'http://darknetlive.com/', 'Date/Time': '2024-04-02 01:30:22'}

Evidence: [operation_dark_web_sample.csv \(Row 4\)](#) | Time: 2024-04-02 01:30:22 | Artifact: web_4 | Confidence: High

Finding: URL visited: {'Artifact Type': 'TSK_WEB_HISTORY', 'Path': 'http://zqktlwi4fecvo6ri.onion/', 'Date/Time': '2024-04-02 02:15:44'}

Evidence: [operation_dark_web_sample.csv \(Row 2\)](#) | Time: 2024-04-02 02:15:44 | Artifact: web_2 | Confidence: High

3.2 Email Analysis

Analysis revealed 2 email message(s) containing relevant correspondence and metadata.

Finding: Email: {'Artifact Type': 'TSK_EMAIL_MESSAGE', 'Date/Time': '2024-04-01 22:30:00', 'Description': 'From: vendor_elite@pro'}

Evidence: [operation_dark_web_sample.csv \(Row 7\)](#) | Time: 2024-04-01 22:30:00 | Artifact: email_7 | Confidence: High

Finding: Email: {'Artifact Type': 'TSK_EMAIL_MESSAGE', 'Date/Time': '2024-04-02 03:05:18', 'Description': 'From: anon@protonmail'}

Evidence: [operation_dark_web_sample.csv \(Row 6\)](#) | Time: 2024-04-02 03:05:18 | Artifact: email_6 | Confidence: High

3.3 Device Connections

1 external storage device(s) identified with connection timestamps and device identifiers.

Finding: USB device: {'Artifact Type': 'TSK_DEVICE_ATTACHED', 'Path': 'E:', 'Date/Time': '2024-04-02 01:00:00', 'Description': 'K'}

Evidence: [operation_dark_web_sample.csv \(Row 8\)](#) | Time: 2024-04-02 01:00:00 | Artifact: usb_8 | Confidence: High

3.4 Deleted Files

File system analysis recovered 2 deleted file(s). Recovery indicates possible deliberate destruction of evidence.

Finding: Deleted: transaction_history_xmr.csv

Evidence: [operation_dark_web_sample.csv \(Row 13\)](#) | Time: 2024-04-02 03:30:00 | Artifact: del_13 | Confidence: High

Finding: Deleted: vendor_contacts_darkweb.txt

Evidence: [operation_dark_web_sample.csv \(Row 14\)](#) | Time: 2024-04-02 03:31:00 | Artifact: del_14 | Confidence: High

3.5 User Accounts

1 user account(s) identified providing insight into access patterns.

Finding: Account: {'Artifact Type': 'TSK_OS_ACCOUNT', 'Path': 'C:\\Users\\alex_r', 'Date/Time': '2024-04-02 00:50:00', 'Description': 'User account for alex_r'}

Evidence: operation_dark_web_sample.csv (Row 12) | Time: 2024-04-02 00:50:00 | Artifact: usr_12 | Confidence: Medium

3.6 Search Queries

2 search query record(s) recovered revealing information-seeking behavior.

Finding: Searched: {'Artifact Type': 'TSK_KEYWORD_HIT', 'Path': 'Browser History', 'Date/Time': '2024-04-01 23:00:00', 'Descripti

Evidence: operation_dark_web_sample.csv (Row 16) | Time: 2024-04-01 23:00:00 | Artifact: srch_16 | Confidence: Medium

Finding: Searched: {'Artifact Type': 'TSK_KEYWORD_HIT', 'Path': 'Browser History', 'Date/Time': '2024-04-02 02:15:44', 'Descripti

Evidence: operation_dark_web_sample.csv (Row 15) | Time: 2024-04-02 02:15:44 | Artifact: srch_15 | Confidence: Medium

3.7 Installed Programs

3 installed program(s) identified. May reveal tools used for data exfiltration.

Finding: Program: Electrum Wallet

Evidence: operation_dark_web_sample.csv (Row 11) | Time: 2024-03-15 09:00:00 | Artifact: prog_11 | Confidence: Medium

Finding: Program: VeraCrypt 1.25

Evidence: operation_dark_web_sample.csv (Row 10) | Time: 2024-03-20 11:00:00 | Artifact: prog_10 | Confidence: Medium

Finding: Program: Tor Browser 13.0

Evidence: operation_dark_web_sample.csv (Row 9) | Time: 2024-03-28 20:00:00 | Artifact: prog_9 | Confidence: Medium

4. FORENSIC TIMELINE

Timestamp	Event	Description	Source (Row)	Conf.
2024-03-15 09:00:0	PROGRAM_INSTALLED	Program: Electrum Wallet	operation_dark_web_sample.cs	Medi
2024-03-15 09:00:0	TIMELINE_EVENT	{'Artifact Type': 'TSK_INSTALLED_PROG', 'Program Name': 'Electrum Wallet', 'Path': 'C:\\Program Files\\Electrum\\Electrum-1.24.0\\Electrum-1.24.0.exe', 'Date Installed': '2024-03-15 09:00:0'}	operation_dark_web_sample.cs	Medi
2024-03-20 11:00:0	PROGRAM_INSTALLED	Program: VeraCrypt 1.25	operation_dark_web_sample.cs	Medi
2024-03-20 11:00:0	TIMELINE_EVENT	{'Artifact Type': 'TSK_INSTALLED_PROG', 'Program Name': 'VeraCrypt', 'Path': 'C:\\Program Files\\VeraCrypt\\VeraCrypt-1.25\\VeraCrypt-1.25.exe', 'Date Installed': '2024-03-20 11:00:0'}	operation_dark_web_sample.cs	Medi
2024-03-28 20:00:0	PROGRAM_INSTALLED	Program: Tor Browser 13.0	operation_dark_web_sample.cs	Medi
2024-03-28 20:00:0	TIMELINE_EVENT	{'Artifact Type': 'TSK_INSTALLED_PROG', 'Program Name': 'Tor Browser', 'Path': 'C:\\Program Files\\Tor\\TorBrowser-13.0\\TorBrowser-13.0.exe', 'Date Installed': '2024-03-28 20:00:0'}	operation_dark_web_sample.cs	Medi
2024-04-01 22:30:0	EMAIL	Email: {'Artifact Type': 'TSK_EMAIL_MESSAGE', 'Subject': 'Dark Web Alert', 'From': 'info@darkweb.com', 'To': 'user@example.com', 'Body': 'Urgent: New dark web marketplace identified. Details in the attached file.', 'Date': '2024-04-01 22:30:0'}	operation_dark_web_sample.cs	High
2024-04-01 22:30:0	TIMELINE_EVENT	{'Artifact Type': 'TSK_EMAIL_MESSAGE', 'Subject': 'Dark Web Alert', 'From': 'info@darkweb.com', 'To': 'user@example.com', 'Body': 'Urgent: New dark web marketplace identified. Details in the attached file.', 'Date': '2024-04-01 22:30:0'}	operation_dark_web_sample.cs	Medi
2024-04-01 22:44:1	WEB_ACCESS	URL visited: {'Artifact Type': 'TSK_WEB_HISTORY', 'URL': 'https://darkweb.onion/marketplace', 'Referrer': 'https://darkweb.onion/login', 'User Agent': 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/120.0.0.0 Safari/537.36', 'Date': '2024-04-01 22:44:1'}	operation_dark_web_sample.cs	High
2024-04-01 22:44:1	TIMELINE_EVENT	{'Artifact Type': 'TSK_WEB_HISTORY', 'URL': 'https://darkweb.onion/marketplace', 'Referrer': 'https://darkweb.onion/login', 'User Agent': 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/120.0.0.0 Safari/537.36', 'Date': '2024-04-01 22:44:1'}	operation_dark_web_sample.cs	Medi
2024-04-01 23:00:0	SEARCH_QUERY	Searched: {'Artifact Type': 'TSK_KEYWORD_HIT', 'Query': 'dark web marketplace', 'Results': 1, 'Date': '2024-04-01 23:00:0'}	operation_dark_web_sample.cs	Medi
2024-04-01 23:00:0	TIMELINE_EVENT	{'Artifact Type': 'TSK_KEYWORD_HIT', 'Query': 'dark web marketplace', 'Results': 1, 'Date': '2024-04-01 23:00:0'}	operation_dark_web_sample.cs	Medi
2024-04-01 23:58:0	WEB_ACCESS	URL visited: {'Artifact Type': 'TSK_WEB_HISTORY', 'URL': 'https://darkweb.onion/marketplace', 'Referrer': 'https://darkweb.onion/login', 'User Agent': 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/120.0.0.0 Safari/537.36', 'Date': '2024-04-01 23:58:0'}	operation_dark_web_sample.cs	High
2024-04-01 23:58:0	TIMELINE_EVENT	{'Artifact Type': 'TSK_WEB_HISTORY', 'URL': 'https://darkweb.onion/marketplace', 'Referrer': 'https://darkweb.onion/login', 'User Agent': 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/120.0.0.0 Safari/537.36', 'Date': '2024-04-01 23:58:0'}	operation_dark_web_sample.cs	Medi
2024-04-02 00:50:0	USER_ACCOUNT	Account: {'Artifact Type': 'TSK_OS_ACCOUNT', 'Username': 'user', 'Password': 'P@ssw0rd123!', 'Last Login': '2024-04-02 00:50:0'}	operation_dark_web_sample.cs	Medi
2024-04-02 00:50:0	TIMELINE_EVENT	{'Artifact Type': 'TSK_OS_ACCOUNT', 'Username': 'user', 'Password': 'P@ssw0rd123!', 'Last Login': '2024-04-02 00:50:0'}	operation_dark_web_sample.cs	Medi
2024-04-02 01:00:0	USB_EVENT	USB device: {'Artifact Type': 'TSK_DEVICE_ATTACHED', 'Device Name': 'USB Flash Drive', 'Path': 'E:\\', 'Date': '2024-04-02 01:00:0'}	operation_dark_web_sample.cs	High
2024-04-02 01:00:0	TIMELINE_EVENT	{'Artifact Type': 'TSK_DEVICE_ATTACHED', 'Device Name': 'USB Flash Drive', 'Path': 'E:\\', 'Date': '2024-04-02 01:00:0'}	operation_dark_web_sample.cs	Medi
2024-04-02 01:30:2	WEB_ACCESS	URL visited: {'Artifact Type': 'TSK_WEB_HISTORY', 'URL': 'https://darkweb.onion/marketplace', 'Referrer': 'https://darkweb.onion/login', 'User Agent': 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/120.0.0.0 Safari/537.36', 'Date': '2024-04-02 01:30:2'}	operation_dark_web_sample.cs	High
2024-04-02 01:30:2	TIMELINE_EVENT	{'Artifact Type': 'TSK_WEB_HISTORY', 'URL': 'https://darkweb.onion/marketplace', 'Referrer': 'https://darkweb.onion/login', 'User Agent': 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/120.0.0.0 Safari/537.36', 'Date': '2024-04-02 01:30:2'}	operation_dark_web_sample.cs	Medi
2024-04-02 02:15:4	WEB_ACCESS	URL visited: {'Artifact Type': 'TSK_WEB_HISTORY', 'URL': 'https://darkweb.onion/marketplace', 'Referrer': 'https://darkweb.onion/login', 'User Agent': 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/120.0.0.0 Safari/537.36', 'Date': '2024-04-02 02:15:4'}	operation_dark_web_sample.cs	High
2024-04-02 02:15:4	SEARCH_QUERY	Searched: {'Artifact Type': 'TSK_KEYWORD_HIT', 'Query': 'dark web marketplace', 'Results': 1, 'Date': '2024-04-02 02:15:4'}	operation_dark_web_sample.cs	Medi
2024-04-02 02:15:4	TIMELINE_EVENT	{'Artifact Type': 'TSK_WEB_HISTORY', 'URL': 'https://darkweb.onion/marketplace', 'Referrer': 'https://darkweb.onion/login', 'User Agent': 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/120.0.0.0 Safari/537.36', 'Date': '2024-04-02 02:15:4'}	operation_dark_web_sample.cs	Medi
2024-04-02 02:15:4	TIMELINE_EVENT	{'Artifact Type': 'TSK_KEYWORD_HIT', 'Query': 'dark web marketplace', 'Results': 1, 'Date': '2024-04-02 02:15:4'}	operation_dark_web_sample.cs	Medi
2024-04-02 03:05:1	EMAIL	Email: {'Artifact Type': 'TSK_EMAIL_MESSAGE', 'Subject': 'Dark Web Alert', 'From': 'info@darkweb.com', 'To': 'user@example.com', 'Body': 'Urgent: New dark web marketplace identified. Details in the attached file.', 'Date': '2024-04-02 03:05:1'}	operation_dark_web_sample.cs	High
2024-04-02 03:05:1	TIMELINE_EVENT	{'Artifact Type': 'TSK_EMAIL_MESSAGE', 'Subject': 'Dark Web Alert', 'From': 'info@darkweb.com', 'To': 'user@example.com', 'Body': 'Urgent: New dark web marketplace identified. Details in the attached file.', 'Date': '2024-04-02 03:05:1'}	operation_dark_web_sample.cs	Medi
2024-04-02 03:30:0	FILE_DELETED	Deleted: transaction_history_xmr.csv	operation_dark_web_sample.cs	High
2024-04-02 03:30:0	TIMELINE_EVENT	{'Artifact Type': 'TSK_RECYCLE_BIN', 'File Name': 'transaction_history_xmr.csv', 'Path': 'C:\\Users\\user\\AppData\\Local\\Electrum\\', 'Date Deleted': '2024-04-02 03:30:0'}	operation_dark_web_sample.cs	Medi
2024-04-02 03:31:0	FILE_DELETED	Deleted: vendor_contacts_darkweb.txt	operation_dark_web_sample.cs	High
2024-04-02 03:31:0	TIMELINE_EVENT	{'Artifact Type': 'TSK_RECYCLE_BIN', 'File Name': 'vendor_contacts_darkweb.txt', 'Path': 'C:\\Users\\user\\AppData\\Local\\Tor\\', 'Date Deleted': '2024-04-02 03:31:0'}	operation_dark_web_sample.cs	Medi

5. EVIDENCE REFERENCES

Complete list of 30 evidence artifact(s) cited in this report:

#	Artifact ID	Source File	Row	Timestamp	Confide
1	web_5	operation_dark_web_sam	5	2024-04-01 22:44:1	High
2	web_3	operation_dark_web_sam	3	2024-04-01 23:58:0	High
3	web_4	operation_dark_web_sam	4	2024-04-02 01:30:2	High
4	web_2	operation_dark_web_sam	2	2024-04-02 02:15:4	High
5	email_7	operation_dark_web_sam	7	2024-04-01 22:30:0	High
6	email_6	operation_dark_web_sam	6	2024-04-02 03:05:1	High
7	usb_8	operation_dark_web_sam	8	2024-04-02 01:00:0	High
8	del_13	operation_dark_web_sam	13	2024-04-02 03:30:0	High
9	del_14	operation_dark_web_sam	14	2024-04-02 03:31:0	High
10	usr_12	operation_dark_web_sam	12	2024-04-02 00:50:0	Medi
11	srch_16	operation_dark_web_sam	16	2024-04-01 23:00:0	Medi
12	srch_15	operation_dark_web_sam	15	2024-04-02 02:15:4	Medi
13	prog_11	operation_dark_web_sam	11	2024-03-15 09:00:0	Medi
14	prog_10	operation_dark_web_sam	10	2024-03-20 11:00:0	Medi
15	prog_9	operation_dark_web_sam	9	2024-03-28 20:00:0	Medi
16	tl_11	operation_dark_web_sam	11	2024-03-15 09:00:0	Medi
17	tl_10	operation_dark_web_sam	10	2024-03-20 11:00:0	Medi
18	tl_9	operation_dark_web_sam	9	2024-03-28 20:00:0	Medi
19	tl_7	operation_dark_web_sam	7	2024-04-01 22:30:0	Medi
20	tl_5	operation_dark_web_sam	5	2024-04-01 22:44:1	Medi
21	tl_16	operation_dark_web_sam	16	2024-04-01 23:00:0	Medi
22	tl_3	operation_dark_web_sam	3	2024-04-01 23:58:0	Medi
23	tl_12	operation_dark_web_sam	12	2024-04-02 00:50:0	Medi
24	tl_8	operation_dark_web_sam	8	2024-04-02 01:00:0	Medi
25	tl_4	operation_dark_web_sam	4	2024-04-02 01:30:2	Medi
26	tl_2	operation_dark_web_sam	2	2024-04-02 02:15:4	Medi
27	tl_15	operation_dark_web_sam	15	2024-04-02 02:15:4	Medi
28	tl_6	operation_dark_web_sam	6	2024-04-02 03:05:1	Medi
29	tl_13	operation_dark_web_sam	13	2024-04-02 03:30:0	Medi
30	tl_14	operation_dark_web_sam	14	2024-04-02 03:31:0	Medi

6. CONCLUSIONS

The examination of Operation Dark Web yielded 15 recoverable artifacts with 0 high-confidence events identified.

All conclusions are based solely on the digital evidence recovered and analyzed. The forensic examiner makes no determination of guilt or innocence.

7. RECOMMENDATIONS

1. Further investigation of identified URLs and online accounts is recommended.
2. Recovered deleted files should be subjected to detailed content analysis.
3. Email communications should be cross-referenced with known contacts.
4. External storage devices should be located and forensically examined.
5. All recovered artifacts should be reviewed by qualified legal counsel.
6. Chain of custody documentation must be maintained throughout legal proceedings.

8. CHAIN OF CUSTODY

Evidence Item:	operation_dark_web_sample.csv
MD5 Hash:	691c7d3608fbc2ff5c2d699ff43b56ab
Examined By:	Agent K. Patel
Department:	Digital Forensics Lab
Analysis Date:	2026-05-28T11:04:27
Tools Used:	Autopsy Parser v3, RAG Engine v3, Automated Digital Forensics Reporter v3
Hash Verification:	MD5 hash computed and recorded
Notes:	Evidence handled in accordance with forensic best practices and ACPO guidelines.

END OF REPORT — Automated Digital Forensics Reporter v3.0